Lab Task: Blockchain Integrity Verification

Files: `blockchain.py`, `blockheader.py`, `database.py`

Objective

Implement a method named `validate_chain()` in the `Blockchain` class to check whether the blockchain is valid.

Requirements

Your method should:

1. Return `False` if the blockchain is empty.
2. Check that the first block has height 1 and uses `ZERO_HASH` as its previous hash.
3. Check that each block has the correct height and previous-block hash.
4. Recalculate each block's hash and compare it with the stored hash.
5. Check that each block's hash satisfies the mining target (`TARGET`).

Return `True` only if all blocks pass these checks. Otherwise, return `False`.

Important: The existing mining code increments the nonce after calculating the hash. Consider this behavior when recalculating a block's hash.

Demonstration

Show the output for these two cases:

Case 1: Original blockchain

Blockchain Valid: True

Case 2: After modifying a block's nonce

Blockchain Valid: False

Submission

Submit the updated `blockchain.py` file and a screenshot showing the output of both test cases.